

PAPER_559: Buoyancy-Stratified Factorial Geometry — Einstein Tensor and Self-Sourced Field Equations

Unified Quantum Field Framework (UQFF)

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***Context note:** PAPER_554 (CP4 #149) derived the Riemann curvature $R^{\{0r\}}$ and Ricci scalar $R_{\rm scalar}$ for the BSFG metric $A_{\mu\nu}$. This paper takes the next step: forming the Einstein tensor $G_{\mu\nu}$ and establishing the BSFG self-sourced field equations. The central finding — that the BSFG amplification factor $\mathrm{amp} \gg 1$ — shows that the Aether-metric perturbation does not obey the standard Einstein equations, but instead a stronger BSFG self-sourcing relation.*

§1 Abstract

We derive the Einstein tensor $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}A_{\mu\nu}R_{\rm scalar}$ for the Buoyancy-Stratified Factorial Geometry metric $A_{\mu\nu}(r) = g_{\mu\nu} + \varepsilon(r)\delta_{\mu\nu}$, and compare it to the natural Aether source $\kappa_E T_{s00}$ via standard Einstein equations. The key result is:

$$\mathrm{amp} \equiv \frac{G_{00}}{\kappa_E T_{s00}} \approx \frac{18\eta c^4}{8\pi G r^2}$$

At the solar surface: $\mathrm{amp} \approx 1.8 \times 10^4$. This amplification means the BSFG metric perturbation $\varepsilon = \eta T_{s00} \cos(\pi t_n)$ generates curvature geometrically out of proportion to any local stress-energy source — a hallmark of a non-Einstein field theory. The effective cosmological constant is

$\Lambda_{\text{eff}} = \kappa_E \eta T_{s00}/2 \approx 1.3 \times 10^{-45} \text{ m}^{-2}$, some seven orders of magnitude above the observed $\Lambda_{\text{obs}} = 1.1 \times 10^{-52} \text{ m}^{-2}$.

§2 Einstein Tensor of the BSFG Metric

From PAPER_554 (CP4 #149), the non-zero Riemann and Ricci components are:

$$R_{00} \approx \frac{\epsilon}{2} = \frac{6\eta \cos(\pi t_n) C_{\text{num}}}{r^5}$$

$$R_{00} = 3R_{00}, \quad R_{\text{scalar}} = \frac{R_{00}}{A_{00}} + \frac{R_{rr}}{A_{rr}}$$

The Einstein tensor is:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} A_{\mu\nu} R_{\text{scalar}}$$

Step 1. Compute components at leading order in ϵ :

$$G_{00} = R_{00} - \frac{1}{2} A_{00} R_{\text{scalar}} = \frac{3}{2} \epsilon - \frac{1}{2} (1 + \epsilon) \left(\frac{R_{00}}{A_{00}} + \frac{R_{rr}}{A_{rr}} \right)$$

$$G_{rr} = R_{rr} - \frac{1}{2} A_{rr} R_{\text{scalar}}$$

Step 2. At $r = R_{\text{odot}}$, $t_n = 0$:

Quantity	Value
R_{00}	$1.56 \times 10^{-19} \text{ m}^{-2}$
$R_{00} = 3R_{00}$	$4.67 \times 10^{-19} \text{ m}^{-2}$
R_{scalar}	$\approx 3.12 \times 10^{-19} \text{ m}^{-2}$
G_{00}	$\approx 3.11 \times 10^{-19} \text{ m}^{-2}$
G_{rr}	$\approx 3.10 \times 10^{-19} \text{ m}^{-2}$

§3 BSFG Field Equations

Step 3. The natural Aether energy density (from $T_{s00}(r)$, in Pa):

$$T_{s00}(R_{\text{odot}}) = \frac{M_s c^2}{\frac{4}{3} \pi R_{\text{odot}}^3} \approx 1.27 \times 10^{20} \text{ Pa}$$

The Einstein gravitational constant:

$$\kappa_E = \frac{8\pi G}{c^4} \approx 2.07 \times 10^{-43} \frac{\text{m}}{\text{kg}}$$

Step 4. Standard GR would require:

$$G_{00} \stackrel{?}{=} \kappa_E T_{s00} \approx 2.63 \times 10^{-23} \text{ m}^{-2}$$

but the actual BSFG $G_{00} \approx 3.11 \times 10^{-19} \text{ m}^{-2}$. The amplification factor:

$$\boxed{\mathrm{amp}} = \frac{G_{00}}{\kappa_E T_{s00}} \approx \frac{18 \eta c^4}{8\pi G r^2} \approx 1.8 \times 10^4$$

This factor $\sim r^{-2}$ — the curvature amplification grows as we approach the origin.

§4 Effective Cosmological Constant

Step 5. Taking the trace of the BSFG field equation hypothesis $G_{\mu\nu} = \kappa_E \eta T_{s00} A_{\mu\nu}$:

$$\Lambda_{\mathrm{eff}} = \frac{\kappa_E \eta T_{s00}}{2} = \frac{4\pi G \eta T_{s00}}{c^4}$$

At $r = R_{\odot}$:

$$\Lambda_{\mathrm{eff}}(R_{\odot}) = 2.07 \times 10^{-43} \times 1 \times 10^{-22} \times 1.27 \times 10^{20} / 2 \approx 1.3 \times 10^{-45} \text{ m}^{-2}$$

The cosmological ratio:

$$\frac{\Lambda_{\mathrm{eff}}}{\Lambda_{\mathrm{obs}}} = \frac{1.3 \times 10^{-45}}{1.1 \times 10^{-52}} \approx 1.2 \times 10^7$$

Interpretation: The BSFG Aether field carries an effective dark-energy-like contribution seven orders of magnitude above the present cosmological constant — but it is not constant; it scales as $T_{s00}(r) \propto r^{-3}$, averaging to near zero over cosmological volumes.

The effective vacuum energy density:

$$\rho_{\mathrm{vac}}^{\mathrm{eff}} = \frac{\Lambda_{\mathrm{eff}} c^2}{8\pi G}$$

§5 Physical Interpretation

The BSFG metric is defined through $\varepsilon = \eta T_{s00} \cos(\pi t_n)$, an explicit algebraic relation from CP4 #43. This is **not** derived from solving the Einstein equations — it is an imposed geometric structure. The fact that $\mathrm{amp} \gg 1$ confirms:

1. **BSFG is not a solution of the standard Einstein equations** sourced by T_{s00} .
2. The correct BSFG field equation is the constitutive relation $\varepsilon = \eta T_{s00} \cos(\pi t_n)$ itself.
3. The Einstein tensor $G_{\mu\nu}$ serves as a diagnostic: it measures the curvature consequences of this constitutive relation.

4. The amp factor $\approx 18\eta c^4/(8\pi G r^2)$ is purely geometric and grows like c^4/G — the same factor that makes gravity weak compared to other forces.

§6 References

- CP4 #149 — [BSFGriemannCurvatureAetherMetricCalculator](#) — PAPER_554 (Riemann curvature)
- CP4 #43 — Aether metric coupling $\eta = 10^{-22}$, PAPER_392
- CP4 #66 — $T_{s00}(r)$ five-component decomposition, PAPER_416
- CP4 #153 — [BSFGUnificationAtlasTheoremHubCalculator](#) — PAPER_558 (complete BSFG definition)